Paired Associates Learning 5 (PAL5)

Objective

To evaluate the effect of closed loop stimulation on memory during the encoding phase of an associative memory task. Stimulation will be applied in response to low classifier states during encoding.

Note: Prior to running PAL5, the subject will have completed at least one session of PS4 to select optimal stimulation parameters. The PS4 design is described as part of the FR5 task.

Timeline of each Session

- **Lists 1-4:** no stimulation. Used to set the mean and standard deviation for normalization. First list is "practice."
- **Lists 5-26:** PAL5 with 16 Stim lists and 6 NoStim lists, pseudorandomly interleaved such that 8 Stim lists occur for every 3 NoStim lists.

Stimulation parameters in the search space

- *Brain Location (N = 2)*: Bipolar pairs, selected based on lateral temporal anatomy.
- *Amplitude:* Continuous on range [0 A_{safe}].
    - Depths A_{safe} = 1.0 mA
    - Grids A_{safe} = 2.0 mA

Stimulation parameters that are fixed

- Stimulation Train Duration: 500 ms
- Stimulation Train Frequency: 200 Hz
- Stimulation Pulse Width: 300 μs biphasic
- Stimulation Decision: In response to low-classifier states
- Stimulation Timing, Encoding: 2100 to 2600 ms after word-pair onset
- Stimulation Timing, Retrieval: sliding 500ms window, 400-900ms to 3500-4000ms after cue onset

Task phases during which to stimulate during PAL5

- Encoding
- *(Optional)* Flag in config file to potentially enable/disable PAL5 Retrieval stim

Classifier (training and closed-loop decoding)

- Single classifier trained with both encoding and retrieval data
- Minimum required record-only data: 3 PAL1 sessions + 1 FR1 session
    - Encoding decision threshold
        - The median classifier output from encoding period of record-only PAL1 sessions and lists 1-4 of PAL 5 will be used.
- Classifier Input Features
    - Encoding and retrieval data from PAL1 and FR1 sessions
    - Bipolar referenced EEG signals
    - Line noise filtering (4th order band-stop Butterworth filter, 58-62 Hz)

    - Stimulation channels excluded from classifier
    - Stimulation epochs excluded from classifier
    - Morlet wavelet spectral power (wave number = 5; center frequencies = 8 log-spaced frequencies from 6-180 Hz [6.00, 9.75, 15.86, 25.78, 41.90, 68.11, 110.73, 180.00] Hz
    - Log-transform the raw power values
    - Z-score power values within feature (frequency X electrode)
        - Z-score using session mean and standard dev.
    - **Encoding data:** spectral power computed from 300-2000 ms relative to each word presentation event. Mirrored buffers (length 1198 ms).
    - **Retrieval data:** spectral power computed from -600 to -100 ms relative to true or surrogate response times
        - **(1) True Responses:** First vocalized word during retrieval period, could be correct or incorrect response.
        - **(2) Surrogate Responses:** For trials without any responses, a response time will be randomly drawn from the distribution of true responses for that subject.
- Classifier Model
    - L2-penalized logistic regression
    - Penalty parameter: fixed value provided by Data Science and updated as needed.
        - **Current value: .048**

    - Penalty parameter weighting:
        - Class A: $(1/n_A)/((1/n_A + 1/n_B)/2)$
        - Class A: $(1/n_B)/((1/n_B + 1/n_A)/2)$
        - n_A = number of class A observations
        - n_B = number of class B observations

    - Weighting of **PAL1 vs FR1** observations: fixed value provided by Data Science and updated as needed
        - **Current value: 3.0**

    - Weighting of **Encoding/Retrieval** phase observations: fixed value provided by Data Science and updated as needed.
        - **Current value: 4.0**

- Classifier Training Assessment
    - At least 3 PAL1 and 1 FR1 sessions required (minimum 15 lists each)
    - Area under the curve (AUC) computed using Leave-one-session-out (LOSO) cross-validation
    - Significance: The true LOSO AUC will be compared to an AUC distribution (200 samples) generated from classification of label-permuted input data. $P < 0.05$ one-tailed will be used as the significance threshold for continuing to PAL5.

Closed Loop Decoding Timeline

- Collect baseline period data for mean and SD normalization (Lists 1-4)
    - Separate mean and SDs for normalizing the Encoding and Retrieval periods.

    - These distributions will be used to normalize both the encoding and retrieval data in lists 5-26.
- Assess classifier generalization to the current closed-loop session
    - After List 4, compare classifier output from Lists 1-4 to the distribution of classifier output from the record-only sessions
    - Two-sample Kolmogorov-Smirnov test
    - If null hypothesis is rejected ($P < 0.05$), abort experiment. Else, continue with Lists 5-26.
- Encoding Period Events:
    - Word-pair Onset Occurs
        - Decode word interval (300-2000ms)
        - Is classifier output < encoding threshold?
            - Yes: stimulate using parameters supplied by PS4
            - No: wait for next word-pair presentation

Closed Loop Data Logging

- Raw EEG: all encoding and retrieval periods on which classifier output was computed
- Spectral power: the processed (wavelet decomposed, log-transformed, z-score normalized) spectral power patterns that were used as input to the classifier.
- Classifier output: the probability value generated by the model.
- Vector of voice detection state output for all decoded retrieval periods.